

Xinyu (Brian) Guo

GENOMIC FOUNDATION MODELS · CANCER GENOMICS · SCIENTIFIC AI AGENTS

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EDUCATION

Ph.D., Computational Biology & Bioinformatics University of Southern California · 3.95/4.0 · Viterbi Fellow Aug 2022 – 2026

M.S., Biostatistics Johns Hopkins University · 3.97/4.0 · Delta Omega Honor Society Aug 2020 – May 2022

B.A., Mathematics & Computer Science Washington University in St. Louis · 3.89/4.0 · Cum Laude Aug 2018 – May 2020

EXPERIENCE

AI Research Scientist Intern — Abbott Cancer Diagnostics (formerly Exact Sciences) Summer 2026

Genomic AI & precision oncology · San Diego (La Jolla), CA

- Researched DNA foundation models — particularly sequence-to-function models such as AlphaGenome and Enigma — and applied them to an OncoExTra tumor cohort, **rescuing 2,554 of 6,087 variants (42%)** that conventional snpEff annotation had deprioritized, and ranking them into a wet-lab handoff.
- Built **OncoS2F**, a scientific agent that translates natural-language research requests into executable Enigma, Evo2, and Carbon workflows, automating sequence-context construction, batch embeddings, feature extraction, model routing, and provenance across **13 hosted genomic model endpoints**.
- Built FFPE artifact-risk models on 87,307 raw somatic SNVs reaching **ROC-AUC 0.923** under leave-one-patient-out evaluation, against 0.677 (FFPErase), 0.517 (FFPolish), and 0.509 (FIFA) on the same cohort; scaled the pipeline to an 18.6M-row SEQC2 benchmark across five callers.
- Extended the same sequence-to-function models from SNVs to structural rearrangements by building a reusable **SV→REF/ALT encoder** that renders deletions, duplications, inversions, breakends, and fusions as model-ready sequence, and curated an **83,048-fusion training corpus** across 1,065 cancer types with evidence tiers and leakage-safe splits for supervised fusion modeling.

Graduate Researcher — University of Southern California Aug 2022 – Present

Ph.D. dissertation research, biological foundation models · Advisor: Dr. Liang Chen

- Developed a **cross-modality foundation model** integrating DNA sequence, RNA expression, and spatial transcriptomics to identify tissue-specific disease-risk regions, built around explainability so individual genetic variants map to interpretable spatial microenvironments rather than opaque scores.
- Designed a **self-supervised contrastive learning** method with GNN/GAT architectures for spatial transcriptomics, enabling integration with large-scale multi-omics cohorts across scRNA-seq, scATAC-seq, and spatial modalities for genetic risk evaluation.
- Proposed a fusion architecture using weighted message-passing GNNs and attention to project GWAS/TWAS signals into spatial embeddings, bridging population-scale genetics with tissue-resolved measurement.
- Identified **trans-acting NMD-QTLs** associated with nonsense-mediated mRNA decay across human tissues, and benchmarked AlphaGenome and Enigma for spatially resolved variant-to-gene perturbation prediction.
- Benchmarked methods for **cell-cycle phase inference** from single-cell RNA-seq, establishing which approaches stay reliable across datasets and where they break down — first-author study in *Briefings in Bioinformatics*.

Graduate Researcher — Johns Hopkins University Oct 2020 – Aug 2022

Statistical genetics · Advisors: Dr. Nilanjan Chatterjee & Dr. Diptavo Dutta

- Developed **CSTWAS**, a subset-based cross-tissue meta-analysis method for transcriptome-wide association studies that raises power over standard single- and multi-tissue tests while identifying which tissues are actually "active" for a gene-trait signal, improving both power and interpretability.
- Performed transcriptome- and proteome-wide association studies at bulk and single-cell resolution to identify genes linked to **renal cell carcinoma**, and contributed to follow-on work partitioning polygenic risk scores to identify cancer-related proteins.
- Modeled GWAS summary statistics jointly with single-cell ATAC-seq to interpret disease etiology at cellular rather than tissue resolution.

Graduate Researcher — Johns Hopkins University Feb 2021 – Aug 2022

Multi-omics & cancer therapy resistance · Advisors: Dr. Hongkai Ji & Dr. Eneda Toska

- Built a comprehensive **RNA-seq, ChIP-seq, and ATAC-seq pipeline** supporting differential analysis, gene-set enrichment, and motif discovery, which became shared infrastructure across several lab projects.

- Elucidated how transcriptional and epigenetic regulators mediate hormone-driven tumor resistance to therapeutics, with the chromatin modifier **KMT2D** and its methylation by SMYD2 as the central axis.
- Co-authored four resistance-mechanism papers across breast, prostate, and melanoma models in *Cell Reports*, *Oncogene*, and *Cancer Research*.

Research Assistant — Washington University in St. Louis

Jun – Aug 2020

Predictive clinical informatics · Advisor: Dr. Andrew Michelson

- Built Lasso and logistic-regression models predicting respiratory failure in COVID-19 patients, and **implemented them in the Barnes-Jewish Hospital EHR** — research code that reached live clinical use during the pandemic.
- Enabled early identification of high-risk patients, supporting timely ICU resource deployment under acute bed and ventilator constraints.

SELECTED PROJECTS

- Engineered and trained a **560M-parameter decoder-only LLM from scratch** 2025 — BPE tokenization, data pipeline, and distributed PyTorch DDP across 8×A100 80GB GPUs, managing the full lifecycle including mixed-precision training and gradient-synchronization tuning for cluster throughput.
- **TAL-Mamba** 2025 – 2026 · lead developer — state-space Temporal Action Localization model for long-horizon sequence understanding (2+ hours), with efficient state management over massive temporal datasets, reducing memory usage **68%** versus Transformer baselines.
- **Track Mamba** 2025 – 2026 · principal architect — linear-complexity SSM tracker for real-time small-object tracking on edge hardware, **40% smaller** and **50% faster** than TrackNet baselines at higher precision–recall.
- **Legal Patent Copilot** 2024 — retrieval agent for prior-art discovery over 10M+ documents using FAISS indexing and fine-tuned reranking layers, cutting search latency **42%** with improved recall.

SELECTED PUBLICATIONS

1. Guo, X., Chen, L. From G1 to M: a comparative study of methods for identifying cell cycle phases. *Briefings in Bioinformatics* (2024).
2. Guo, X., Chatterjee, N., Dutta, D. Subset-based method for cross-tissue transcriptome-wide association studies improves power and interpretability. *Human Genetics and Genomics Advances* (2024).
3. Dutta, D., Guo, X., Winter, T.D., et al. Transcriptome- and proteome-wide association studies identify genes associated with renal cell carcinoma. *The American Journal of Human Genetics* (2024).
4. Bianchi, A., De Castro Silva, I., ... Guo, X., et al. Cell-autonomous Cxcl1 sustains tolerogenic circuitries and stromal inflammation via neutrophil-derived TNF in pancreatic cancer. *Cancer Discovery* (2023).
5. Blawski, R., Vokshi, B.H., Guo, X., et al. Methylation of the chromatin modifier KMT2D by SMYD2 contributes to therapeutic response in hormone-dependent breast cancer. *Cell Reports* (2024).
6. Kittane, S., Ladewig, E., ... Guo, X., et al. The histone methyltransferase KMT2D mediates subtype-specific transcriptional regulation, lineage plasticity, and therapeutic response in castration-resistant prostate cancer. *Cancer Research* (2024, 2025).
7. Jagirdar, K., Portuallo, M.E., ... Guo, X., et al. ERK hyperactivation as a unified mechanism of escape in intrinsic and acquired CDK4/6-inhibitor resistance in acral lentiginous melanoma. *Oncogene* (2024).
8. Dutta, D., Zhang, J., Guo, X., et al. Leveraging and partitioning polygenic risk scores to identify cancer-related proteins. *medRxiv* (2025, preprint).

TECHNICAL EXPERTISE

AI / ML Genomic foundation models (AlphaGenome, Enigma, Evo2, Carbon, Borzoi, Enformer, DNABERT) · transformers · state-space models (Mamba) · GNN/GAT · self-supervised & contrastive learning · LLM pretraining · agentic systems & tool use · RAG / vector retrieval · leakage-safe evaluation

COMP. BIOLOGY Cancer genomics & precision oncology · variant-effect prediction (SNV, SV, fusion) · regulatory genomics & splicing · GWAS/TWAS · single-cell and spatial transcriptomics · RNA-seq / ChIP-seq / ATAC-seq · NGS somatic calling, FFPE artifacts, VCF · OncoKB, CIViC

ENGINEERING Python · R · C++ · SQL · Bash · TypeScript · PyTorch (DDP/FSDP), PyTorch Geometric, Hugging Face, JAX, scikit-learn, CUDA · FastAPI · React · Nextflow / WDL · DuckDB, Parquet · Docker, Git

CLOUD / SCALE AWS SageMaker GPU inference · Bedrock · HealthOmics · EC2, ECS, S3 · petabyte-scale sequence data on Linux/HPC · multi-GPU (A100) distributed training